

M-Estimation

1.1 Motivation

Discussion 1.1. Consider an M-estimator $\hat{\boldsymbol{\theta}}_n$ obtained by solving:

$$\boldsymbol{\Psi}_n(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^n \boldsymbol{\psi}(\mathbf{z}_i; \boldsymbol{\theta}) \stackrel{\text{Set}}{=} \mathbf{0}$$

Recall that the influence function expansion for $\hat{\boldsymbol{\theta}}_n$ is:

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{A}(\boldsymbol{\theta}_0)^{-1} \boldsymbol{\psi}(\mathbf{z}_i; \boldsymbol{\theta}_0) + o_p(1)$$

where:

$$\mathbf{A}(\boldsymbol{\theta}_0) = -\mathbb{E} \left\{ \frac{\partial \boldsymbol{\psi}(\mathbf{z}_i; \boldsymbol{\theta})}{\partial \boldsymbol{\theta}'} \right\}_{\boldsymbol{\theta}=\boldsymbol{\theta}_0},$$

and that the limiting distribution of $\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0)$ is:

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{\mathcal{L}} N\{\mathbf{0}, \mathbf{A}(\boldsymbol{\theta}_0)^{-1} \mathbf{B}(\boldsymbol{\theta}_0) \mathbf{A}(\boldsymbol{\theta}_0)^{-T}\},$$

where:

$$\mathbf{B}(\boldsymbol{\theta}_0) = \mathbb{E}\{\boldsymbol{\psi}(\mathbf{z}_i; \boldsymbol{\theta}) \otimes \boldsymbol{\psi}(\mathbf{z}_i; \boldsymbol{\theta})\}_{\boldsymbol{\theta}=\boldsymbol{\theta}_0}.$$

Let $\tilde{\boldsymbol{\theta}}_n$ denote the estimator obtained by solving the perturbed estimating equations:

$$\tilde{\boldsymbol{\Psi}}_n(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^n \boldsymbol{\psi}(\mathbf{z}_i; \boldsymbol{\theta}) \cdot \omega_i \stackrel{\text{Set}}{=} \mathbf{0},$$

where (ω_i) are IID random weights, with $\mathbb{E}(\omega_i) = 0$ and $\mathbb{E}(\omega_i^2) = 1$, that are generated independent of the data.

The **score bootstrap** approximates the distribution of $\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0)$ by that of:

$$\sqrt{n}(\tilde{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{A}(\hat{\boldsymbol{\theta}}_n)^{-1} \boldsymbol{\psi}(\mathbf{z}_i; \hat{\boldsymbol{\theta}}_n) \cdot \omega_i$$

Conditional on the observed data, the perturbed influence function has expectation:

$$\mathbb{E}\{\sqrt{n}(\tilde{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n)\} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{A}(\hat{\boldsymbol{\theta}}_n)^{-1} \boldsymbol{\psi}(\mathbf{z}_i; \hat{\boldsymbol{\theta}}_n) \cdot \mathbb{E}(\omega_i) = \mathbf{0},$$

and variance:

$$\begin{aligned} \mathbb{V}\{\sqrt{n}(\tilde{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n)\} &= \frac{1}{n} \sum_{i=1}^n \mathbf{A}(\hat{\boldsymbol{\theta}}_n)^{-1} \boldsymbol{\psi}(\mathbf{z}_i; \hat{\boldsymbol{\theta}}_n) \otimes \boldsymbol{\psi}(\mathbf{z}_i; \hat{\boldsymbol{\theta}}_n) \mathbf{A}(\hat{\boldsymbol{\theta}}_n)^{-T} \cdot \mathbb{E}(\omega_i^2) \\ &= \mathbf{A}(\hat{\boldsymbol{\theta}}_n)^{-1} \left\{ \frac{1}{n} \sum_{i=1}^n \boldsymbol{\psi}(\mathbf{z}_i; \hat{\boldsymbol{\theta}}_n) \otimes \boldsymbol{\psi}(\mathbf{z}_i; \hat{\boldsymbol{\theta}}_n) \right\} \mathbf{A}(\hat{\boldsymbol{\theta}}_n)^{-T} \xrightarrow{p} \mathbf{A}_0^{-1} \mathbf{B}_0 \mathbf{A}_0^{-T}. \end{aligned}$$

Consequently, $\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0)$ and $\sqrt{n}(\tilde{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n)$ converge to the same limiting distribution (see Kline & Santos 2012). ♠

1.2 Procedure

1. Estimate $\hat{\theta}_n$ satisfying $\Psi_n(\hat{\theta}_n) = \mathbf{0}$.

i. Calculate the individual score contributions evaluated at $\hat{\theta}_n$:

$$\hat{\psi}_i = \psi(\mathbf{z}_i; \hat{\theta}_n).$$

ii. Estimate the information matrix $\hat{\mathbf{A}}_n$:

$$\hat{\mathbf{A}}_n = -\frac{1}{n} \sum_{i=1}^n \left\{ \frac{\partial \psi(\mathbf{z}_i; \boldsymbol{\theta})}{\partial \boldsymbol{\theta}'} \right\}_{\boldsymbol{\theta}=\hat{\theta}_n}.$$

2. For $b = 1, \dots, B$:

i. Generate the perturbation weights $(\omega_i^{(b)})$.

ii. Calculate the perturbed score:

$$\mathcal{U}_n^{(b)} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \hat{\mathbf{A}}_n^{-1} \hat{\psi}_i \cdot \omega_i^{(b)}.$$

Note that for hypothesis testing perturbations should be performed under H_0 .

3. Approximate the distribution of $\sqrt{n}(\hat{\theta}_n - \boldsymbol{\theta})$ by that of $\mathcal{U}_n^{(b)}$.

Examples

2.1 Huber Estimator

Example 2.1 (Confidence Intervals). Suppose (Y_i) are a random sample from a symmetric distribution with location θ_0 . Recall that the Huber estimator $\hat{\theta}_n$ is a solution to the estimating equation:

$$\Psi_n(\theta) = \sum_{i=1}^n \psi_\tau(Y_i - \theta) \stackrel{\text{Set}}{=} 0,$$

where $\psi_\tau(u)$ is the Huber function with threshold τ :

$$\psi_\tau(u) = \begin{cases} -\tau & u < -\tau, \\ u & -\tau \leq u \leq \tau, \\ \tau & u > \tau. \end{cases}$$

The influence function expansion for $\hat{\theta}_n$ is:

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{1}{\sqrt{n}} A_0^{-1} \sum_{i=1}^n \psi_\tau(Y_i - \theta_0) + o_p(1),$$

where A_0 is the information matrix:

$$A_0 = -\mathbb{E} \left\{ \frac{\partial \psi(Y_i - \theta)}{\partial \theta} \right\} = -\mathbb{E} \{ \dot{\psi}_\tau(Y_i - \theta_0) \}, \quad \dot{\psi}_\tau(u) = \begin{cases} 1, & |u| \leq \tau, \\ 0, & |u| > \tau. \end{cases}$$

Asymptotically:

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{\mathcal{L}} N(0, A_0^{-1} B_0 A_0^{-1}),$$

where:

$$B_0 = \mathbb{E} \{ \psi_\tau^2(Y_i - \theta_0) \}.$$

The goal of perturbation is to approximate the distribution of the pivotal quantity:

$$Z_n = \sqrt{n} \frac{(\hat{\theta}_n - \theta_0)}{\sigma_n} = \frac{1}{\sqrt{n}} \sigma_n^{-1} A(\theta_0)^{-1} \sum_{i=1}^n \psi_\tau(Y_i - \theta_0) + o_p(1).$$

Towards this, obtain the individual score contributions $\hat{\psi}_i = \psi_\tau(Y_i - \hat{\theta}_n)$, and the empirical information:

$$\hat{A}_n = \frac{1}{n} \sum_{i=1}^n \dot{\psi}_\tau(Y_i - \hat{\theta}_n).$$

For $b \in \{1, \dots, B\}$, generate the perturbation weights $(\omega_i^{(b)})$ and calculate:

$$Z_n^{(b)} = \frac{1}{\sqrt{n}} (\sigma_n^{(b)})^{-1} \hat{A}_n^{-1} \sum_{i=1}^n \hat{\psi}_i \cdot \omega_i^{(b)},$$

where the standard error is:

$$(\sigma_n^{(b)})^2 = \frac{\hat{B}_n^{(b)}}{\hat{A}_n^2}, \quad \hat{B}_n^{(b)} = \frac{1}{n} \sum_{i=1}^n \hat{\psi}_i^2 \cdot (\omega_i^{(b)})^2.$$

Obtain critical values (ζ_*, ζ^*) from the quantiles of $(Z_n^{(b)})$ such that:

$$P \left(\zeta_* \leq \sqrt{n} \frac{(\hat{\theta}_n - \theta)}{\sigma_n} \leq \zeta^* \right) = 1 - \alpha.$$

Rearranging provides the confidence interval for θ :

$$\text{CI} = \left(\theta : \hat{\theta}_n - \zeta^* \frac{\sigma_n}{\sqrt{n}} \leq \theta \leq \hat{\theta}_n - \zeta_* \frac{\sigma_n}{\sqrt{n}} \right).$$



Example 2.2 (Testing). The Wald statistic for evaluating the $H_0 : \theta = \theta_0$ is:

$$T_W = n \frac{(\hat{\theta} - \theta_0)^2}{\sigma_n^2}. \quad (2.1)$$

The goal of perturbation is to approximate the distribution of T_W under H_0 . Towards this, for $b \in \{1, \dots, B\}$, generate the perturbation weights $(\omega_i^{(b)})$ and calculate:

$$\mathcal{U}_n^{(b)} = \frac{1}{\sqrt{n}} \tilde{A}_n^{-1} \sum_{i=1}^n \tilde{\psi}_i \cdot \omega_i^{(b)}, \quad (\tilde{\sigma}_n^{(b)})^2 = \frac{\tilde{B}_n^{(b)}}{\tilde{A}_n^2}, \quad \tilde{B}_n^{(b)} = \frac{1}{n} \sum_{i=1}^n \tilde{\psi}_i^2 \cdot (\omega_i^{(b)})^2.$$

Construct the perturbation test statistics:

$$T_W^{(b)} = \frac{(\mathcal{U}_n^{(b)})^2}{(\tilde{\sigma}_n^{(b)})^2}.$$

A p-value assessing $H_0 : \theta = \theta_0$ is given by:

$$p = \frac{1}{B+1} \left\{ 1 + \sum_{b=1}^B I(T^{(b)} \geq T_{\text{obs}}) \right\}.$$



2.2 Linear Models

Example 2.3 (Confidence Intervals). Consider the model:

$$Y_i = \mathbf{x}_i' \boldsymbol{\beta} + \epsilon_i,$$

where $\mathbb{E}(\epsilon_i | \mathbf{x}_i) = 0$ and $\mathbb{V}(\epsilon_i | \mathbf{x}_i) = \sigma_i^2$.

The OLS estimator $\hat{\boldsymbol{\beta}}_n$ is a solution to the estimating equation:

$$\boldsymbol{\Psi}_n(\boldsymbol{\beta}) = \mathbf{X}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \sum_{i=1}^n \mathbf{x}_i(Y_i - \mathbf{x}_i' \boldsymbol{\beta}) \stackrel{\text{Set}}{=} \mathbf{0}.$$

Identify the estimating function as:

$$\boldsymbol{\psi}_i = \mathbf{x}_i(Y_i - \mathbf{x}_i' \boldsymbol{\beta}).$$

The influence function expansion for $\hat{\boldsymbol{\beta}}_n$ is:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}_0) = \frac{1}{\sqrt{n}} \mathbf{A}_0^{-1} \mathbf{X}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + o_p(1),$$

where $\mathbf{A}_0$ is the unit information for $\boldsymbol{\beta}$:

$$\mathbf{A}_0 = \text{plim}_{n \rightarrow \infty} \frac{1}{n} \mathcal{I}_{\boldsymbol{\beta}\boldsymbol{\beta}'} = \text{plim}_{n \rightarrow \infty} \frac{1}{n} \mathbf{X}' \mathbf{X}.$$

Asymptotically:

$$\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{\mathcal{L}} N(\mathbf{0}, \mathbf{A}_0^{-1} \mathbf{B}_0 \mathbf{A}_0^{-T}),$$

where:

$$\mathbf{B}_0 = \mathbb{E}\{\boldsymbol{\psi}(\mathbf{x}_i; \beta_0) \otimes \boldsymbol{\psi}(\mathbf{x}_i; \beta_0)\} = \mathbb{E}\{(Y_i - \mathbf{x}_i' \beta_0)^2 \mathbf{x}_i \otimes \mathbf{x}_i\}.$$

The goal of perturbation is to approximate the distribution of the pivotal quantity:

$$\mathbf{Z}_n = \sqrt{n} \boldsymbol{\Sigma}_n^{-1/2} (\hat{\beta}_n - \beta_0) = \frac{1}{\sqrt{n}} \boldsymbol{\Sigma}_n^{-1/2} \mathbf{A}_0^{-1} \mathbf{X}' (\mathbf{y} - \mathbf{X} \beta_0) + o_p(1),$$

where $\boldsymbol{\Sigma}_n$ is a consistent estimator for the asymptotic variance:

$$\boldsymbol{\Sigma}_n = \hat{\mathbf{A}}_n^{-1} \hat{\mathbf{B}}_n \hat{\mathbf{A}}_n^{-T}.$$

The empirical information:

$$\hat{\mathbf{A}}_n = \frac{1}{n} \mathbf{X}' \mathbf{X}.$$

For $b \in \{1, \dots, B\}$, generate the perturbation weights and calculate:

$$\mathbf{Z}_n^{(b)} = \frac{1}{\sqrt{n}} (\boldsymbol{\Sigma}_n^{(b)})^{-1/2} \hat{\mathbf{A}}_n^{-1} \mathbf{X}' \boldsymbol{\Omega}^{(b)} \hat{\mathbf{e}}_n,$$

where $\hat{\mathbf{e}}_n = (\mathbf{y} - \mathbf{X} \hat{\beta}_n)$, $\boldsymbol{\Omega}^{(b)} = \text{diag}(\omega_i^{(b)})$, $\boldsymbol{\Sigma}_n^{(b)} = \hat{\mathbf{A}}_n^{-1} \hat{\mathbf{B}}_n^{(b)} \hat{\mathbf{A}}_n^{-T}$, and:

$$\hat{\mathbf{B}}_n^{(b)} = \frac{1}{n} \mathbf{X}' \boldsymbol{\Omega}^{(b)} \text{diag}(\hat{e}_n^2) \boldsymbol{\Omega}^{(b)} \mathbf{X}.$$

Obtain critical values (ζ_*, ζ^*) from the quantiles of $(\mathbf{Z}_n^{(b)})$ such that:

$$P\left(\zeta_* \leq \sqrt{n} \boldsymbol{\Sigma}_n^{-1/2} (\hat{\beta}_n - \beta) \leq \zeta^*\right) = 1 - \alpha.$$

Rearranging provides the confidence interval for β :

$$\text{CI} = \left(\beta : \hat{\beta}_n - n^{-1/2} \boldsymbol{\Sigma}_n^{1/2} \zeta_* \leq \beta \leq \hat{\beta}_n - n^{-1/2} \boldsymbol{\Sigma}_n^{1/2} \zeta^*\right).$$



Example 2.4 (Testing). Consider the model:

$$Y_i = \mathbf{x}_i' \boldsymbol{\alpha} + \mathbf{z}_i' \boldsymbol{\beta} + \epsilon_i,$$

where $\mathbb{E}(\epsilon_i | \mathbf{x}_i, \mathbf{z}_i) = 0$ and $\mathbb{V}(\epsilon_i | \mathbf{x}_i, \mathbf{z}_i) = \sigma_i^2$. Suppose $H_0 : \boldsymbol{\beta} = \boldsymbol{\beta}_0$ is of interest. Let $\tilde{\boldsymbol{\alpha}}_n$ denote a solution to the OLS estimating equation:

$$\mathcal{U}_\alpha(\boldsymbol{\alpha}, \boldsymbol{\beta}_0) = \mathbf{X}'(\mathbf{y} - \mathbf{X}\boldsymbol{\alpha} - \mathbf{Z}\boldsymbol{\beta}_0) \stackrel{\text{Set}}{=} \mathbf{0}.$$

Consider the distribution of the score statistic:

$$\mathcal{U}_\beta(\tilde{\boldsymbol{\alpha}}_n, \boldsymbol{\beta}_0) = \mathbf{Z}'(\mathbf{y} - \mathbf{X}\tilde{\boldsymbol{\alpha}}_n - \mathbf{Z}\boldsymbol{\beta}_0).$$

Take the Taylor expansions of $\mathcal{U}_\alpha$ and $\mathcal{U}_\beta$ about the true $\boldsymbol{\alpha}_0$:

$$\begin{aligned} \mathbf{0} &= \mathcal{U}_\alpha(\tilde{\boldsymbol{\alpha}}_n, \boldsymbol{\beta}_0) = \mathcal{U}_\alpha(\boldsymbol{\alpha}_0, \boldsymbol{\beta}_0) - \mathcal{I}_{\alpha\alpha'}(\tilde{\boldsymbol{\alpha}}_n - \boldsymbol{\alpha}_0) + \mathcal{O}_p(1), \\ \mathcal{U}_\beta(\tilde{\boldsymbol{\alpha}}_n, \boldsymbol{\beta}_0) &= \mathcal{U}_\beta(\boldsymbol{\alpha}_0, \boldsymbol{\beta}_0) - \mathcal{I}_{\beta\alpha'}(\tilde{\boldsymbol{\alpha}}_n - \boldsymbol{\alpha}_0) + \mathcal{O}_p(1), \end{aligned}$$

where $\mathcal{I}$ denotes the information matrix, and the remainder is bounded in probability by assumption. Substitute the first expansion into the second to obtain:

$$\begin{aligned} \mathcal{U}_\beta(\tilde{\boldsymbol{\alpha}}_n, \boldsymbol{\beta}_0) &= \mathcal{U}_\beta(\boldsymbol{\alpha}_0, \boldsymbol{\beta}_0) - \mathcal{I}_{\beta\alpha'} \mathcal{I}_{\alpha\alpha'}^{-1} \mathcal{U}_\alpha(\boldsymbol{\alpha}_0, \boldsymbol{\beta}_0) + \mathcal{O}_p(1) \\ &= \left(-\mathcal{I}_{\beta\alpha'} \mathcal{I}_{\alpha\alpha'}^{-1}, \mathbf{I} \right) \begin{pmatrix} \mathcal{U}_\alpha(\boldsymbol{\alpha}_0, \boldsymbol{\beta}_0) \\ \mathcal{U}_\beta(\boldsymbol{\alpha}_0, \boldsymbol{\beta}_0) \end{pmatrix} + \mathcal{O}_p(1). \end{aligned}$$

Define the following structures:

$$\mathbf{C} = \left(-\mathcal{I}_{\beta\alpha'} \mathcal{I}_{\alpha\alpha'}^{-1}, \mathbf{I} \right), \quad \mathcal{U}_\theta(\boldsymbol{\alpha}_0, \boldsymbol{\beta}_0) = \begin{pmatrix} \mathcal{U}_\alpha(\boldsymbol{\alpha}_0, \boldsymbol{\beta}_0) \\ \mathcal{U}_\beta(\boldsymbol{\alpha}_0, \boldsymbol{\beta}_0) \end{pmatrix}, \quad \boldsymbol{\Xi} = (\mathbf{X}, \mathbf{Z}).$$

The score statistic for $\boldsymbol{\beta}$ is expressible as:

$$\begin{aligned} \frac{1}{\sqrt{n}} \mathcal{U}_\beta(\tilde{\boldsymbol{\alpha}}_n, \boldsymbol{\beta}_0) &= \frac{1}{\sqrt{n}} \mathbf{C} \mathcal{U}_\theta(\boldsymbol{\alpha}_0, \boldsymbol{\beta}_0) + o_p(1) \\ &= \frac{1}{\sqrt{n}} \mathbf{C} \boldsymbol{\Xi}' (\mathbf{y} - \mathbf{X}\boldsymbol{\alpha}_0 - \mathbf{Z}\boldsymbol{\beta}_0) + o_p(1). \end{aligned}$$

By the standard central limit theorem:

$$\frac{1}{\sqrt{n}} \mathcal{U}_\beta(\tilde{\boldsymbol{\alpha}}_n, \boldsymbol{\beta}_0) \xrightarrow{\mathcal{L}} N(\mathbf{0}, \boldsymbol{\Sigma}_0),$$

where:

$$\boldsymbol{\Sigma}_0 = \text{plim}_{n \rightarrow \infty} \frac{1}{n} \mathbf{C} \boldsymbol{\Xi}' \{ \text{diag}(\sigma_i^2) \} \boldsymbol{\Xi} \mathbf{C}'.$$

A score statistic for evaluating $H_0 : \beta = \beta_0$ is:

$$T_S = \frac{1}{n} \mathcal{U}_\beta(\tilde{\alpha}_n, \beta_0)' \{ \mathbf{C} \Xi' \text{diag}(\sigma_i^2) \Xi \mathbf{C}' \}^{-1} \mathcal{U}_\beta(\tilde{\alpha}_n, \beta_0).$$

The goal of perturbation is to calculate the distribution of T_S under H_0 . Towards this, for $b \in \{1, \dots, B\}$, generate the perturbation weights and calculate:

$$\begin{aligned} \mathcal{U}_\beta^{(b)} &= \mathbf{C} \Xi' \Omega^{(b)} \tilde{\mathbf{e}}_n, \\ T_S^{(b)} &= \frac{1}{n} \mathcal{U}_\beta^{(b)'} \{ \mathbf{C} \Xi' \Omega^{(b)} \text{diag}(\tilde{\mathbf{e}}_n^2) \Omega^{(b)} \Xi \mathbf{C}' \}^{-1} \mathcal{U}_\beta^{(b)}, \end{aligned}$$

where $\tilde{\mathbf{e}}_n = (\mathbf{y} - \mathbf{X} \tilde{\alpha}_n - \mathbf{Z} \beta_0)$ and $\Omega^{(b)} = \text{diag}(\omega_i^{(b)})$.

A p-value assessing $H_0 : \beta = \beta_0$ is given by:

$$p = \frac{1}{B+1} \left\{ 1 + \sum_{b=1}^B I(T_S^{(b)} \geq T_{\text{obs}}) \right\}.$$



2.3 Generalized Linear Models

Example 2.5 (Confidence Intervals). Consider a GLM with linear predictor:

$$\eta_i = \mathbf{x}_i' \beta.$$

The maximum likelihood estimator $\hat{\beta}_n$ is a solution to the score equations:

$$\mathcal{U}_\beta = \mathbf{X}' \mathbf{W} \Delta (\mathbf{y} - \boldsymbol{\mu}), \quad \Delta = \text{diag}\{\dot{g}(\mu_i)\}, \quad \mathbf{W} = \text{diag}\left\{ \frac{1}{\phi \nu(\mu_i) \dot{g}^2(\mu_i)} \right\}.$$

Identify the estimating function as:

$$\boldsymbol{\psi}_i = \mathbf{x}_i \frac{(Y_i - \mathbf{x}_i' \beta)}{\phi \nu(\mu_i) \dot{g}(\mu_i)}.$$

The influence function expansion for $\hat{\beta}_n$ is:

$$\sqrt{n}(\hat{\beta}_n - \beta_0) = \frac{1}{\sqrt{n}} \mathbf{A}_0^{-1} \mathcal{U}_\beta(\beta_0) + o_p(1),$$

where $\mathbf{A}_0$ is the unit information for β :

$$\mathbf{A}_0 = \text{plim}_{n \rightarrow \infty} \frac{1}{n} \mathcal{I}_{\beta\beta'} = \text{plim}_{n \rightarrow \infty} \frac{1}{n} \mathbf{X}' \mathbf{W} \mathbf{X}.$$

Asymptotically:

$$\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{\mathcal{L}} N(\mathbf{0}, \mathbf{A}_0^{-1} \mathbf{B}_0 \mathbf{A}_0^{-T}).$$

Recall that:

$$\mathbb{V}(Y_i - \mathbf{x}_i' \boldsymbol{\beta}) = \phi \nu(\mu_i),$$

therefore:

$$\mathbf{B}_0 = \mathbb{E}\{\boldsymbol{\psi}(\mathbf{x}_i; \boldsymbol{\beta}_0) \otimes \boldsymbol{\psi}(\mathbf{x}_i; \boldsymbol{\beta}_0)\} = \mathbb{E}\left\{\frac{\mathbf{x}_i \otimes \mathbf{x}_i}{\phi \nu(\mu_i) \dot{g}^2(\mu_i)}\right\} = \text{plim}_{n \rightarrow \infty} \frac{1}{n} \mathbf{X}' \mathbf{W} \mathbf{X}.$$

The goal of perturbation is to approximate the distribution of the pivotal quantity:

$$\mathbf{Z}_n = \sqrt{n} \boldsymbol{\Sigma}_n^{-1/2} (\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}_0) = \frac{1}{\sqrt{n}} \boldsymbol{\Sigma}_n^{-1/2} \mathbf{A}_0^{-1} \mathbf{X}' \mathbf{W} \boldsymbol{\Delta} (\mathbf{y} - \boldsymbol{\mu}_0) + o_p(1),$$

where $\boldsymbol{\Sigma}_n$ is a consistent estimator for the asymptotic variance:

$$\boldsymbol{\Sigma}_n = \hat{\mathbf{A}}_n^{-1} \hat{\mathbf{B}}_n \hat{\mathbf{A}}_n^{-T}.$$

The empirical information is:

$$\hat{\mathbf{A}}_n = \frac{1}{n} \mathbf{X}' \hat{\mathbf{W}} \mathbf{X}.$$

For $b \in \{1, \dots, B\}$, generate the perturbation weights and calculate:

$$\mathbf{Z}_n^{(b)} = \frac{1}{\sqrt{n}} (\boldsymbol{\Sigma}_n^{(b)})^{-1/2} \mathbf{A}_0^{-1} \mathbf{X}' \mathbf{W} \boldsymbol{\Delta} \boldsymbol{\Omega}^{(b)} \hat{\mathbf{e}},$$

where $\hat{\mathbf{e}}_n = (\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}_n)$, $\boldsymbol{\Omega}^{(b)} = \text{diag}(\omega_i^{(b)})$, $\boldsymbol{\Sigma}_n^{(b)} = \hat{\mathbf{A}}_n^{-1} \hat{\mathbf{B}}_n^{(b)} \hat{\mathbf{A}}_n^{-T}$, and:

$$\hat{\mathbf{B}}_n^{(b)} = \frac{1}{n} \mathbf{X}' \mathbf{W} \boldsymbol{\Delta} \boldsymbol{\Omega}^{(b)} \text{diag}(\hat{\mathbf{e}}_n^2) \boldsymbol{\Omega}^{(b)} \boldsymbol{\Delta} \mathbf{W} \mathbf{X}.$$

Obtain critical values $(\boldsymbol{\zeta}_*, \boldsymbol{\zeta}^*)$ from the quantiles of $(\mathbf{Z}_n^{(b)})$ such that:

$$P\left(\boldsymbol{\zeta}_* \leq \sqrt{n} \boldsymbol{\Sigma}_n^{-1/2} (\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}) \leq \boldsymbol{\zeta}^*\right) = 1 - \alpha.$$

Rearranging provides the confidence interval for $\boldsymbol{\beta}$:

$$\text{CI} = \left(\boldsymbol{\beta} : \hat{\boldsymbol{\beta}}_n - n^{-1/2} \boldsymbol{\Sigma}_n^{1/2} \boldsymbol{\zeta}^* \leq \boldsymbol{\beta} \leq \hat{\boldsymbol{\beta}}_n - n^{-1/2} \boldsymbol{\Sigma}_n^{1/2} \boldsymbol{\zeta}_*\right).$$



Example 2.6 (Testing). Consider a GLM with linear predictor:

$$\eta_i = \mathbf{x}_i' \boldsymbol{\alpha} + \mathbf{z}_i' \boldsymbol{\beta}.$$

Suppose $H_0 : \boldsymbol{\beta} = \boldsymbol{\beta}_0$ is of interest. Let $\tilde{\boldsymbol{\alpha}}$ denote a solution to the estimating equation:

$$\mathcal{U}_{\boldsymbol{\alpha}}(\boldsymbol{\alpha}, \boldsymbol{\beta}_0) = \mathbf{Z}' \mathbf{W} \boldsymbol{\Delta} \{\mathbf{y} - \boldsymbol{\mu}(\boldsymbol{\alpha}, \boldsymbol{\beta}_0)\}.$$

Following the same derivation used for linear models:

$$\begin{aligned}\frac{1}{\sqrt{n}}\mathcal{U}_\beta(\tilde{\alpha}_n, \beta_0) &= \frac{1}{\sqrt{n}}\mathbf{C}\mathcal{U}_\theta(\alpha_0, \beta_0) + o_p(1) \\ &= \frac{1}{\sqrt{n}}\mathbf{C}\Xi'\mathbf{W}\Delta(\mathbf{y} - \boldsymbol{\mu}_0) + o_p(1).\end{aligned}$$

A score statistic for evaluating $H_0 : \beta = \beta_0$ is:

$$T_S = \frac{1}{n}\mathcal{U}_\beta(\tilde{\alpha}_n, \beta_0)' \{ \Xi'\mathbf{W}\Delta \text{Var}(\mathbf{y}) \Delta \mathbf{W}\Xi \}^{-1} \mathcal{U}_\beta(\tilde{\alpha}_n, \beta_0).$$

The goal of perturbation is to calculate the distribution of T_S under H_0 . Towards this, for $b \in \{1, \dots, B\}$, generate the perturbation weights and calculate:

$$\begin{aligned}\mathcal{U}_\beta^{(b)} &= \mathbf{C}\Xi'\mathbf{W}\Delta\Omega^{(b)}\tilde{\mathbf{e}}_n, \\ T_S^{(b)} &= \frac{1}{n}\mathcal{U}_\beta^{(b)'} \{ \mathbf{C}\Xi'\mathbf{W}\Delta\Omega^{(b)}\text{diag}(\tilde{\mathbf{e}}_n^2)\Omega^{(b)}\Delta\mathbf{W}\Xi\mathbf{C}' \}^{-1} \mathcal{U}_\beta^{(b)},\end{aligned}$$

where $\tilde{\mathbf{e}}_n = (\mathbf{y} - \mathbf{X}\tilde{\alpha}_n - \mathbf{Z}\beta_0)$ and $\Omega^{(b)} = \text{diag}(\omega_i^{(b)})$.

A p-value assessing $H_0 : \beta = \beta_0$ is given by:

$$p = \frac{1}{B+1} \left\{ 1 + \sum_{b=1}^B I(T_S^{(b)} \geq T_{\text{obs}}) \right\}.$$

Under the canonical link, the preceding are simplified by the identity $\mathbf{W}\Delta = \mathbf{I}$. ♠

References

- Kline & Santos. *A Score Based Approach to Wild Bootstrap Inference*. Journal of Econometric Methods (2012).